

# LEARNER GUIDE

|                               |                                                                                                             |
|-------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Unit Standard Title:</b>   | Describe, apply, analyse and calculate shape and motion in 2-and 3-dimensional space in different contexts. |
| <b>Unit Standard No:</b>      | 9013                                                                                                        |
| <b>Unit Standard Credits:</b> | 4                                                                                                           |
| <b>NQF Level:</b>             | 3                                                                                                           |

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This material has been compiled based on the **UNIT STANDARD 119465: WRITE/ PRESENT SIGN TEXTS FOR A RANGE OF COMMUNICATIVE CONTEXTS**

The program has been accredited by SASSETA in the Republic of South Africa.

Material compiled by:

Training Dynamics Africa

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## 1. Introduction

This learning program is part of a complete qualification. For more details concerning the complete qualification please contact your service provider or alternatively the office of the SASSETA

## 2. Purpose of this learning program

This unit standard is designed to provide credits towards the mathematical literacy requirements of the NQF at level 3. The essential purposes of the mathematical literacy requirements are that, as the learner progresses with confidence through the levels, the learner will grow in:

- ♦ An insightful use of mathematics in the management of the needs of everyday living to become a self-managing person.
- ♦ An understanding of mathematical applications that provides insight into the learner's present and future occupational experiences and so develop into a contributing worker.
- ♦ The ability to voice a critical sensitivity to the role of mathematics in a democratic society and so become a participating citizen.

People credited with this unit standard are able to:

- ♦ Measure, estimate, and calculate physical quantities in practical situations relevant to the adult in life or the workplace.
- ♦ Explore describe and represent, interpret and justify geometrical relationships and conjectures to solve problems in two and three dimensional geometrical situations.

### 2.1 Target group

This program is compiled for the following **target group**:

- ♦ Security members
- ♦ South African Defense Force members
- ♦ South African Police Force members
- ♦ Correctional Services
- ♦ Individuals who wishes to complete the NQF level 3 National Certificate in security practices.

## 3. Standards and qualifications

It is important that you as the learner understand and realize that this training program is not presented in isolation, but that it was developed based upon nationally recognized standards known as unit standards.

**REMEMBER.....**

Unit standards are the “building blocks” of qualifications. All qualifications are plotted on the National Qualifications Framework (NQF).

#### 4. Assessments

In order to assess whether a learner can actually demonstrate the desired outcomes, assessment criteria are included in the unit standard. Each outcome has its own set of assessment criteria.

**REMEMBER.....**

**The assessment criteria describe the evidence that is needed that will show that the learner has demonstrated the outcome correctly.**

Kindly refer to the unit standard attached hereto for the assessment criteria.

It is of utmost importance that the learner fully understands the assessment criteria as listed in the unit standard, as it is the only way in which the learner will know what he will be assessed against.

The **final or summative assessment** is the most important aspect of this training program. It is during this process that the learner will be declared competent or not yet competent.

The learner will know exactly how he will be assessed, and when and where he will be assessed. All of these details must be obtained from the training provider where the learner enrolled for his program.

#### 5. Range statements

**REMEMBER.....**

Also included in the unit standard are the range statements in support of the assessment criteria.

**The range statements indicate detailed requirements of the assessment criteria.**

#### 6. The learning guide

The learner guide is included in this material under various learning units. The learner guide has been designed in such a manner that the learner is guided in a logical way through the learning material and requirements of the unit standard.

On completion of this skill program, the learner will be assessed against the assessment criteria as stipulated in the unit standard.

Each of the modules consists of learner outcomes to be achieved during training and development interventions, work related experience or at your own time. The learner outcomes will enable you to achieve the required level of confidence and competence in order to undergo the summative (final evaluations) assessment.

The learner needs to assess his own knowledge and skill throughout the training process by completing the learner workbook. With the completion of the workbook you start collecting evidence to proof your competence. The learner workbook will be assessed (FORMATIVE ASSESSMENT)

**TAKE NOTE**

The learner guide will remain the property of the learner once the LEARNING PROGRAM has been completed.

**TAKE NOTE**

- The best results will be obtained if you start with Study Unit 1 in the Learner Guide, and work your way through it, study unit by study unit.
- All learning outcomes are vital and must be studied (and exercised for all practical requirements) thoroughly to ensure that enough evidence for all specific outcomes is generated.

## **7. Learning assumed to be in place**

It is assumed that a learner which registers for this program have completed a program in communication at NQF Level 3 or equivalent.

It is imperative to understand that this program are only one of the building blocks to complete Skills program 1 The complete list of unit standards for Skills program appears on the next page.

## 7.1 Security program matrix

## UNIT STANDARDS:

|             | ID                     | UNIT STANDARD TITLE                                                                                                                                       | LEVEL   | CREDITS |
|-------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------|
| Core        | <a href="#">244176</a> | Use security equipment                                                                                                                                    | Level 2 | 2       |
| Core        | <a href="#">114941</a> | Apply knowledge of HIV/AIDS to a specific business sector and a workplace                                                                                 | Level 3 | 4       |
| Core        | <a href="#">244184</a> | Apply legal aspects in a security environment                                                                                                             | Level 3 | 8       |
| Core        | <a href="#">244177</a> | Conduct a security patrol in an area of responsibility                                                                                                    | Level 3 | 7       |
| Core        | <a href="#">117705</a> | Demonstrate knowledge of the Firearms Control Act 2000 (Act No 60 of 2000) applicable to possessing a firearm                                             | Level 3 | 3       |
| Core        | <a href="#">246694</a> | Explain the requirements for becoming a security service provider                                                                                         | Level 3 | 4       |
| Core        | <a href="#">244182</a> | Give evidence in court                                                                                                                                    | Level 3 | 4       |
| Core        | <a href="#">244179</a> | Handle complaints and problems                                                                                                                            | Level 3 | 6       |
| Core        | <a href="#">244181</a> | Perform hand over and take over responsibilities                                                                                                          | Level 3 | 2       |
| Core        | <a href="#">244189</a> | Conduct access and egress control                                                                                                                         | Level 4 | 7       |
| Core        | <a href="#">242825</a> | Conduct evacuations and emergency drills                                                                                                                  | Level 4 | 4       |
| Core        | <a href="#">11505</a>  | Identify, handle and defuse security related conflict                                                                                                     | Level 4 | 12      |
| Fundamental | <a href="#">119472</a> | Accommodate audience and context needs in oral/signed communication                                                                                       | Level 3 | 5       |
| Fundamental | <a href="#">9010</a>   | Demonstrate an understanding of the use of different number bases and measurement units and an awareness of error in the context of relevant calculations | Level 3 | 2       |
| Fundamental | <a href="#">9013</a>   | Describe, apply, analyse and calculate shape and motion in 2- and 3-dimensional space in different contexts                                               | Level 3 | 4       |
| Fundamental | <a href="#">119457</a> | Interpret and use information from texts                                                                                                                  | Level 3 | 5       |
| Fundamental | <a href="#">9012</a>   | Investigate life and work related problems using data and probabilities                                                                                   | Level 3 | 5       |
| Fundamental | <a href="#">119467</a> | Use language and communication in occupational learning programmes                                                                                        | Level 3 | 5       |
| Fundamental | <a href="#">7456</a>   | Use mathematics to investigate and monitor the financial aspects of personal, business and national issues                                                | Level 3 | 5       |
| Fundamental | <a href="#">119465</a> | Write/present/sign texts for a range of communicative contexts                                                                                            | Level 3 | 5       |
| Elective    | <a href="#">117889</a> | Demonstrate an understanding of Victim Empowerment                                                                                                        | Level 2 | 6       |
| Elective    | <a href="#">12484</a>  | Perform basic fire fighting                                                                                                                               | Level 2 | 4       |
| Elective    | <a href="#">13912</a>  | Apply knowledge of self and team in order to develop a plan to enhance team performance                                                                   | Level 3 | 5       |
| Elective    | <a href="#">244178</a> | Apply neighbourhood watch principles                                                                                                                      | Level 3 | 5       |
| Elective    | <a href="#">113852</a> | Apply occupational health, safety and environmental principles                                                                                            | Level 3 | 10      |
| Elective    | <a href="#">243188</a> | Care for a service dog                                                                                                                                    | Level 3 | 15      |
| Elective    | <a href="#">116534</a> | Carry out basic first aid treatment in the workplace                                                                                                      | Level 3 | 2       |

**UNIT STANDARD 9013: Describe, apply, analyse and calculate shape and motion in 2-and 3-dimensional space in different contexts.**

|          |                        |                                                                                      |         |    |
|----------|------------------------|--------------------------------------------------------------------------------------|---------|----|
| Elective | <a href="#">113909</a> | Coach a team member in order to enhance individual performance in work environment   | Level 3 | 5  |
| Elective | <a href="#">114942</a> | Describe how to manage reactions arising from a traumatic event                      | Level 3 | 2  |
| Elective | <a href="#">243190</a> | Handle a trained service dog to deter crime                                          | Level 3 | 8  |
| Elective | <a href="#">119649</a> | Handle and use a handgun                                                             | Level 3 | 2  |
| Elective | <a href="#">119651</a> | Handle and use a manually operated rifle or carbine                                  | Level 3 | 2  |
| Elective | <a href="#">119650</a> | Handle and use a self-loading rifle or carbine                                       | Level 3 | 2  |
| Elective | <a href="#">119652</a> | Handle and use a shotgun                                                             | Level 3 | 2  |
| Elective | <a href="#">114979</a> | Operate a computer workstation in a business environment                             | Level 3 | 2  |
| Elective | <a href="#">244335</a> | Conduct security at an event                                                         | Level 4 | 5  |
| Elective | <a href="#">252174</a> | Demonstrate an understanding of security measures within a port security environment | Level 4 | 4  |
| Elective | <a href="#">246693</a> | Explain the requirements for safeguarding a national key point                       | Level 4 | 4  |
| Elective | <a href="#">123515</a> | Handle and use a handgun for business purposes                                       | Level 4 | 3  |
| Elective | <a href="#">123519</a> | Handle and use a manually operated rifle or carbine for business purposes            | Level 4 | 3  |
| Elective | <a href="#">123511</a> | Handle and use a self loading rifle or carbine for business                          | Level 4 | 3  |
| Elective | <a href="#">123514</a> | Handle and use a shotgun for business purposes                                       | Level 4 | 3  |
| Elective | <a href="#">11985</a>  | Inspect, drive and maintain an official vehicle                                      | Level 4 | 3  |
| Elective | <a href="#">11513</a>  | Operate effectively within a specified control room environment                      | Level 4 | 15 |

## **8. Preparation of summative assessment**

After you have worked through the **LEARNER GUIDE** and you are satisfied that all theoretical and practical requirements/ evidence, as stipulated by the UNIT STANDARD, can be met, you need to make your own necessary arrangements with the assessor and/or facilitator. The training establishment will then arrange a suitable date for you to attend your pre-assessment meeting and then arrange for a suitable date to complete the final assessment.

### **8.1 RPL assessment**



The assessment of RPL learners will be conducted in the same way as for those of new learners. The assessment pack is exactly the same and will therefore be used for new learners as well as RPL Learners. It must however be noted that learners who are applying for RPL must provide proof of previous learning and subject related experience prior to the assessment.

This proof or evidence can be in the format of certified copies (certificates) of previous learning programs that have been attended.

All the evidence will be assessed and authenticated before a learner will be allowed to enroll for an RPL program.

## **9. Roles and responsibilities**

### **9.1 Facilitator/instructor**

- Prepare to facilitate.
- Transfer of knowledge and skills

### **9.2 The role of the assessor**

- Plan for the assessment
- Prepare the learner for the assessment

### **9.3 The role of the Learner**

- Prepare for assessment.
- Accept the responsibility of learning.
- Accept the responsibility to produce/deliver evidence.

# STUDY UNIT 1

## Metric System

### LEARNING OUTCOMES

On completion of this study unit the student will....



- ♦ Explain what is meant with accuracy.
- ♦ Explain the term rounding.
- ♦ Do basic calculations using: addition, subtraction, multiplication, division, powers and roots.
- ♦ Apply the measurement system and instruments used in South Africa as set out in this guide.
- ♦ Apply area and volume calculation within an operational environment.
- ♦ Compile a rough sketch of an area or scene of incident.
- ♦ Interpret a plan drawn according to scale.



## 1. Metric system



You must have a good understanding of the metric system in order to measure efficiently. It is very useful to be able to convert from one unit to the other.

Metric units are based on length in metres, mass in grams and volume in litres. The following prefixes are commonly used:

- ♦ Milli: a thousandth (  $\frac{1}{1000}$  )
- ♦ Centi: a hundredth (  $\frac{1}{100}$  )
- ♦ Kilo: one thousand (  $\frac{1000}{1}$  )

Here is a summary of conversions between different units in the metric system:

|    |      | Big units to small units | Small units to big units |
|----|------|--------------------------|--------------------------|
|    |      | →                        | ←                        |
| Km | ↔ m  | × 1 000                  | ÷ 1 000                  |
| m  | ↔ cm | × 100                    | ÷ 100                    |
| m  | ↔ mm | × 1 000                  | ÷ 1 000                  |
| cm | ↔ mm | × 10                     | ÷ 10                     |
| kg | ↔ g  | × 1 000                  | ÷ 1 000                  |
| g  | ↔ mg | × 1 000                  | ÷ 1 000                  |
| kl | ↔ l  | × 1 000                  | ÷ ÷ 1 000                |
| l  | ↔ ml | × 1 000                  | ÷ 1 000                  |

**EXAMPLE:**

Convert 2,35 m to mm. According to the table, multiply by 1 000.

**Solution:**  $2,35 \text{ m} \times 1\,000 = 2\,350 \text{ mm}$

Convert 356 g to kg. According to the table divide by 1 000.

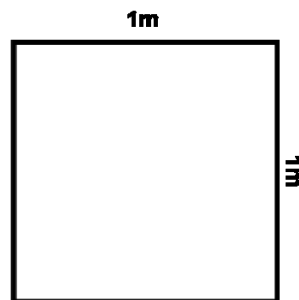
**Solution:**  $356 \text{ g} \div 1\,000 = 0,356 \text{ g}$

When converting between area units, we need to take into account that area is measured in square units.

**EXAMPLE:**

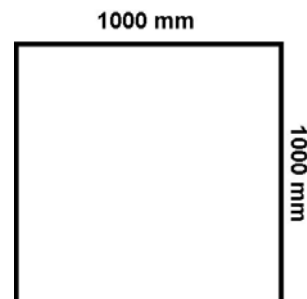
For example, a square with sides the length of one metre has an area of one metre square or  $1 \text{ m}^2$ .

Area:  $1 \text{ m} \times 1 \text{ m} = 1 \text{ m}^2$



However, if we convert the length and width of the square to millimetres: ( $1 \text{ m} = 1\,000 \text{ mm}$ ), we see that the area is 1 000 000 square millimetres ( $\text{mm}^2$ )

Area:  $1\,000 \text{ mm} \times 1\,000 \text{ mm} = 1\,000\,000 \text{ mm}^2$



Hence,  $1 \text{ m}^2 = 1\,000\,000 \text{ mm}^2$

Here is the conversion table for area:

|                 |                   | Big units to small units | Small units to big units |
|-----------------|-------------------|--------------------------|--------------------------|
|                 |                   | →                        | ←                        |
| km <sup>2</sup> | ↔ m <sup>2</sup>  | × 1 000 000              | ÷ 1 000 000              |
| m <sup>2</sup>  | ↔ cm <sup>2</sup> | × 10 000                 | ÷ 10 000                 |
| m <sup>2</sup>  | ↔ mm <sup>2</sup> | × 1 000 000              | ÷ 1 000 000              |
| cm <sup>2</sup> | ↔ mm <sup>2</sup> | × 100                    | ÷ 100                    |

Convert 24,68 m<sup>2</sup> to square centimetres ( cm<sup>2</sup> )

**Solution:** 24,68 m<sup>2</sup> × 10 000 = 246 800 cm<sup>2</sup>

Following the same process for volume, which is measured in cubic units, for example m<sup>3</sup>, km<sup>3</sup>, cm<sup>3</sup>, etc., we get the following conversion table:

|                 |                   | Big units to small units | Small units to big units |
|-----------------|-------------------|--------------------------|--------------------------|
|                 |                   | →                        | ←                        |
| km <sup>3</sup> | ↔ m <sup>3</sup>  | × 1 000 000 000          | ÷ 1 000 000 000          |
| m <sup>3</sup>  | ↔ cm <sup>3</sup> | × 1 000 000              | ÷ 1 000 000              |
| m <sup>3</sup>  | ↔ mm <sup>3</sup> | × 1 000 000 000          | ÷ 1 000 000 000          |
| cm <sup>3</sup> | ↔ mm <sup>3</sup> | × 1000                   | ÷ 1000                   |



Convert 24,68 m<sup>3</sup> to mm<sup>3</sup>

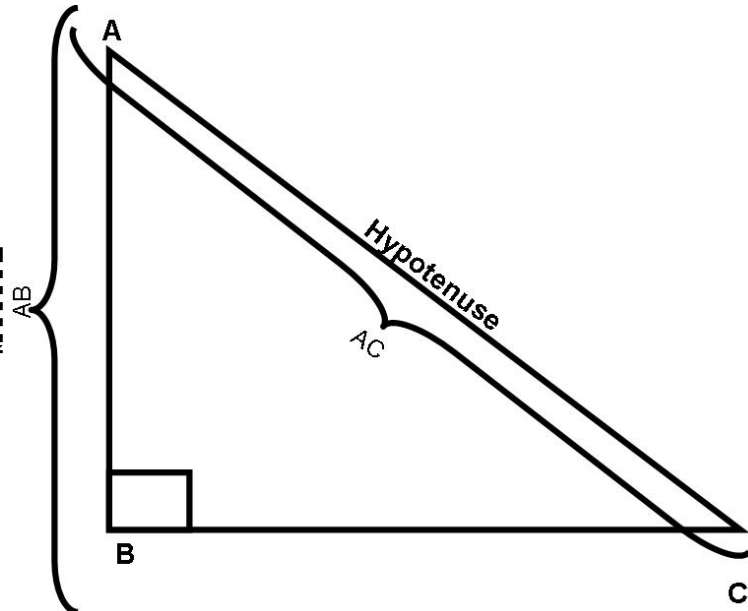
**Solution:** 24,68 m<sup>3</sup> × 1 000 000 000 = 24 680 000 000 mm<sup>3</sup>

## 2. Theorem of Pythagoras

In a right-angled triangle ABC, the side opposite the right angle, AC, is called the hypotenuse. Pythagoras was a Greek mathematician, who discovered that in a right-angled triangle, the square of the hypotenuse will always equal the sum of the squares of the other two sides.

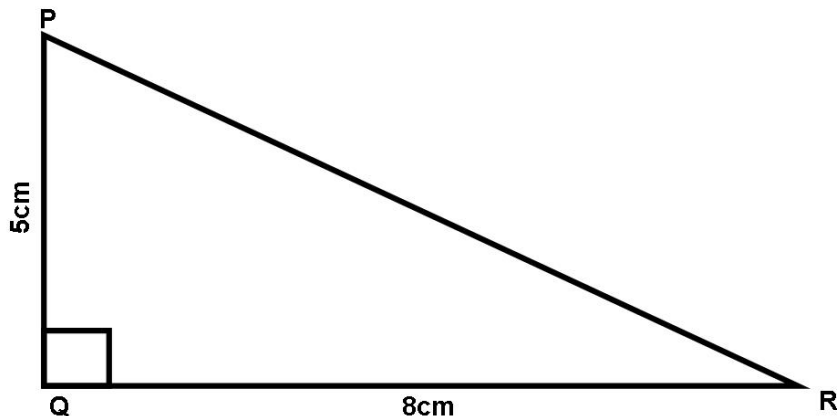
### EXAMPLE:

$$AC^2 = AB^2 + BC^2.$$



### EXAMPLE:

Find the length of side PR of  $\triangle PQR$ , if PQ = 5 cm and QR = 8 cm. Give your answer correct to 1 decimal place.



Solution:

$$\begin{aligned} PR^2 &= PQ^2 + QR^2 \\ &= (5 \text{ cm})^2 + (8 \text{ cm})^2 \\ &= 25 \text{ cm}^2 + 64 \text{ cm}^2 \\ &= 89 \text{ cm}^2 \end{aligned}$$

### 3. Perimeter and Area Formulae of 2-Dimensional Shapes

The **perimeter** is the total *distance* around the outside of a 2-D shape. It is the sum of the lengths of all its sides. You calculate it by adding together all the lengths of a shape.

The area is the amount of surface the 2-D shape covers. It is measured in square units.

- ♦ **Rectangle:** Perimeter =  $2L + 2B$

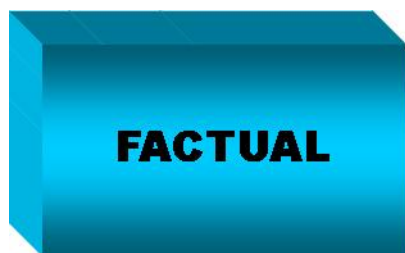
Area =  $L \times B$  (where  $L$  = length,  $B$  = breadth)

- ♦ **Triangle:** Perimeter =  $S + S + S$  (where  $S$  = side)

Area =  $\frac{1}{2} \text{ base} \times \text{perpendicular height} = \frac{1}{2}bh$

- ♦ **Circle:** Circumference (perimeter or distance around circle) =  $2\pi r$  ( $r$  = radius) or  $\pi d$  ( $d$  = diameter)

Area =  $\pi r^2$  ( $\pi \times r \times r$ ) where  $r$  = radius of circle



Pi (  $\pi$  ) is a number that has a value of approximately 3.14159

### 4. Volume and Surface Area of 3-Dimensional Shapes

The **volume** of a 3-D shape is the *space* that it takes up, i.e. its capacity. The volume of any regular 3-D figure (shape) can be calculated as:

Volume = area of base  $\times$  height

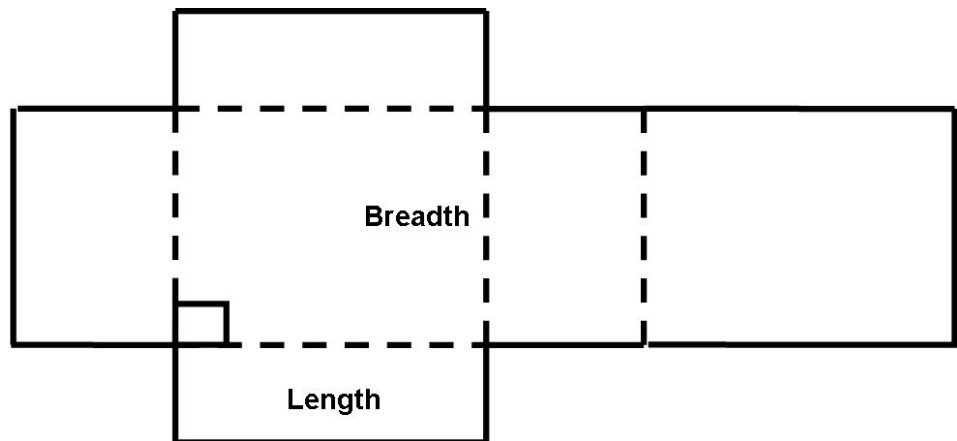
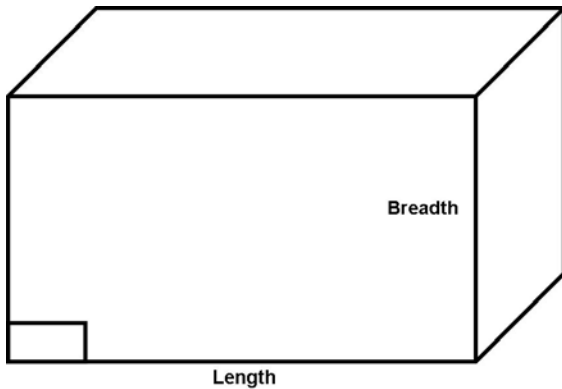
- ♦ **Rectangular prism:** Volume = length  $\times$  breadth  $\times$  height (  $L \times B \times H$  )
- ♦ **Triangular prism:** Volume = (  $\frac{1}{2} \text{ base} \times \text{perpendicular height}$  )  $\times$  length
- ♦ **Cylinder:** Volume =  $\pi r^2 h$

The surface area of a 3-D figure (shape) is the total of the areas of all the faces of the figure. In order to find the total area of a 3-D shape, it has to be adapted to a 2-D shape by determining its net.

## 5. Nets of 3-Dimensional Shapes

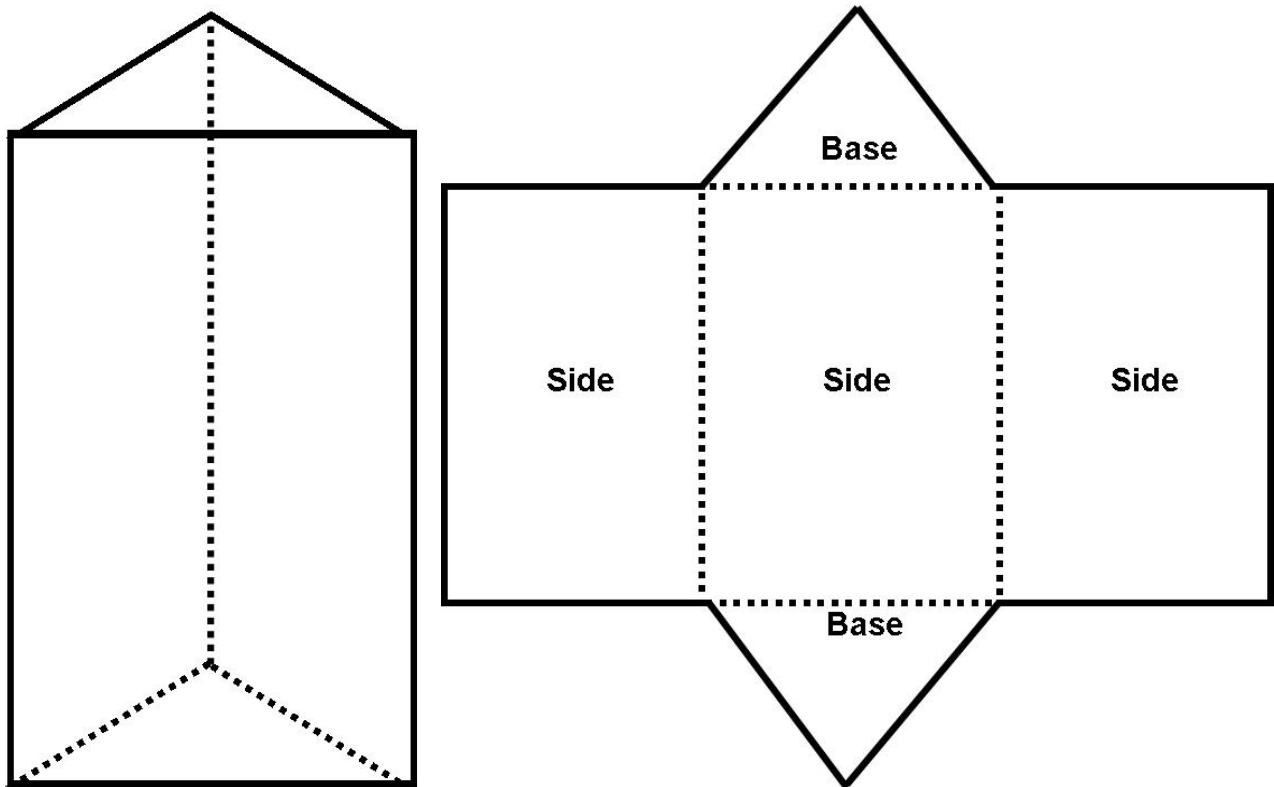
All three-dimensional shapes have *nets*. A net is a flat (2-dimensional) shape that can be folded to form a 3-dimensional shape.

- ♦ **Rectangular prism:** The net of a rectangular prism can vary, but it will always consist of 2 bases (the top and the bottom) and 4 sides - i.e. it will always have 6 faces.



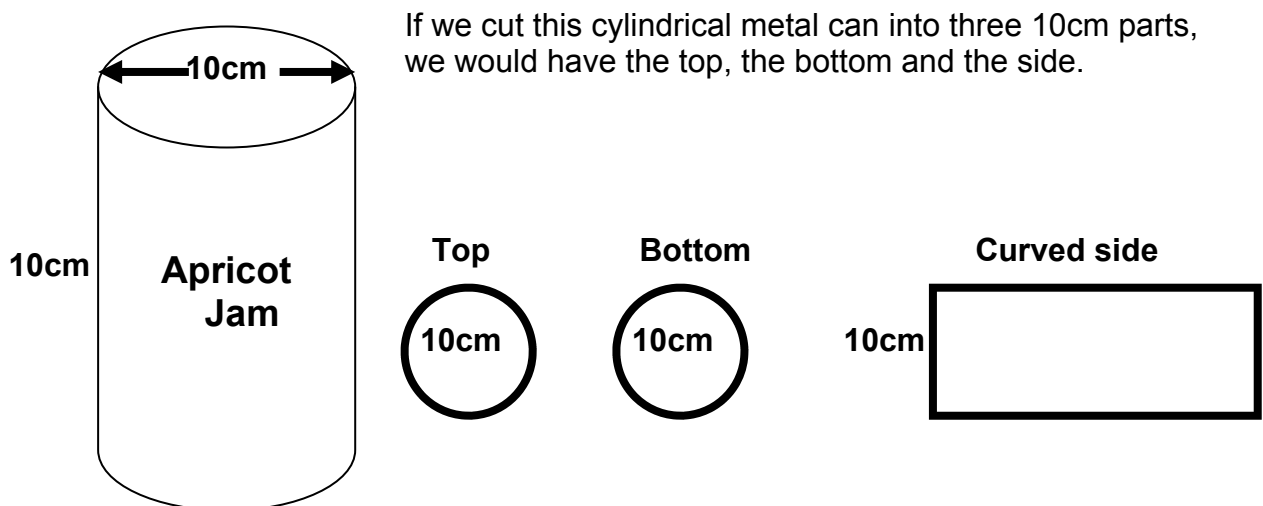
To find the surface area of a rectangular prism, we simply add the areas of each of the 6 faces.

- ♦ **Triangular prism:** Similarly, the net of a triangular prism will always consist of 2 bases (the top and the bottom) and 3 sides - i.e. it will always have 5 faces.



To find the surface area of a triangular prism, we simply add the areas of the 5 faces - 3 rectangles and 2 triangles.

- ♦ **Cylinder:** Let's investigate what the net of a cylinder will look like.



The top and the bottom would be circles, but the side of the can, when opened up, would be a rectangle. The length of the rectangle will be the circumference of the top (or bottom) circle. The breadth of the rectangle will be the height of the can.

This makes it very easy to calculate the surface area of a cylinder:

$$\begin{aligned}\text{Area of top} &= \pi r^2 \\ &= 3,14 \times (5 \text{ cm})^2 \\ &= 78,5 \text{ cm}^2\end{aligned}$$

Therefore: Area of bottom =  $78,5 \text{ cm}^2$

$$\text{Area of side} = L \times B$$

To find the length we need to calculate the circumference of the circle.

$$\begin{aligned}\text{Circumference} &= \pi d \\ &= 3,14 \times 10 \text{ cm} \\ &= 31,4 \text{ cm}\end{aligned}$$

$$\begin{aligned}\text{Therefore: Area of side} &= L \times B \text{ (L is the circumference of the circle)} \\ &= 31,4 \text{ cm} \times 10 \text{ cm} \\ &= 314 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Therefore: Surface area of can} &= 78,5 \text{ cm}^2 + 78,5 \text{ cm}^2 + 314 \text{ cm}^2 \\ &= 471 \text{ cm}^2\end{aligned}$$

## 6. Volume of a cylinder



Remember that the formula for volume is:

$$\text{Volume} = \text{area of base} \times \text{height}$$

A cylinder has a circle for a base. So the base area is  $\pi r^2$  and therefore Volume =  $\pi r^2 h$



$$\begin{aligned}\text{Volume} &= 78,5 \text{ cm}^2 \times 10 \text{ cm} \\ &= 785 \text{ cm}^3\end{aligned}$$

## 7. Speed

Speed is the rate at which an object moves. It is a measure of the change in distance of an object with respect to time.

Simply put, speed =  $\frac{\text{how far an object has travelled}}{\text{how long it takes to get there}}$

**EXAMPLE:**

$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$

**EXAMPLE:**

If a car travels 50 km in 30 minutes, at what speed is it travelling?

Solution: Speed =  $\frac{50 \text{ km}}{30 \text{ min}}$

$$= 50 \text{ km} \div 30 \text{ min} \left( \frac{30}{60} \right)$$

$$= 50 \text{ km} \div 0,5$$

Speed = 100 km/h

**8. Scale drawing**

A scale is the ratio of the length of something on a plan to its actual length on the ground (reality).

A scale of 1:100 means that one unit of length on the drawing represents 100 units of the same length in reality, i.e. 1 cm on the plan represents 100 cm in reality.

So, if something is 4 cm long on the plan, it is really  $4 \times 100$  cm long in reality ,  
i.e. 400 cm  
i.e. 4 m long

**EXAMPLE:**

An architect is drawing a scale drawing of a garage measuring 9 m × 7 m. The scale on his plan is 1:200. What are the dimensions of the garage on the plan?

**Solution:** The architect must *reduce* the dimensions by *dividing them by 200*. Let's first convert from metres to centimetres:

$$9 \text{ m} = 900 \text{ cm} \qquad 7 \text{ m} = 700 \text{ cm}$$

$$900 \text{ cm} \div 200 = 4,5 \text{ cm} \qquad 700 \text{ cm} \div 200 = 3,5 \text{ cm}$$

The architect will draw the garage with dimensions 4,5 cm × 3,5 cm.

When the builder looks at the plan, he will *multiply by 200* to *enlarge* the dimensions to their real life size.



| WORDS/ACRONYMS                  | EXPLANATION OF ACRONYMS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASSESSMENT                      | “A structured process of gathering evidence and making judgments about an individual’s performance in relation to registered national standards and qualifications”                                                                                                                                                                                                                                                                                                                                                                                                   |
| ASSESSMENT CRITERIA             | It describe the evidence that is needed to show that the learner has demonstrated the outcome correctly.<br><br>A list of evidence to be supplied by the candidate as proof of his / her ability to integrate all of the competencies listed earlier in the standard.                                                                                                                                                                                                                                                                                                 |
| COMPETENCE                      | Competence is the ability to do certain things in a range of different environments with knowledge and understanding about what is being done and the effect this has on others. It includes <u>Knowledge, skills, attitudes, values and ethics.</u>                                                                                                                                                                                                                                                                                                                  |
| CREDIT                          | Each unit standard is worth a number of credits which gives an indication of the size and scope of the unit standard, the standard is worth. (1 credit = 10 notional hours of learning)                                                                                                                                                                                                                                                                                                                                                                               |
| CRITICAL CROSSFIELD OUTCOMES    | critical cross field outcomes is to direct educational activities towards development of the learners within a social and economic environment.                                                                                                                                                                                                                                                                                                                                                                                                                       |
| EMBEDDED KNOWLEDGE              | Body of theory essential for understanding and applying the standard. Principles, methods and techniques that would underpin good practice of this unit standard.                                                                                                                                                                                                                                                                                                                                                                                                     |
| EVIDENCE                        | Evidence is a set of information or observations that provides proof of the learner’s competence.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| FORMATIVE ASSESSMENT            | Aims to assist the learner to identify any knowledge/skills gaps when demonstrating his/her current competence. Diagnostic assessment supports learning and progress of the learner.                                                                                                                                                                                                                                                                                                                                                                                  |
| INTEGRATED ASSESSMENT           | Integrated assessment refers to: <ul style="list-style-type: none"> <li>• assessing a number of outcomes together</li> <li>• assessing a number of assessment criteria together</li> <li>• assessing a number of unit standards together</li> <li>• using a combination of assessment methods and instruments for an outcome/s</li> <li>• collecting naturally occurring evidence in the workplace</li> <li>• Acquiring evidence from other sources such as supervisor’s reports, testimonials, portfolio of work previously done, logbooks, journals etc.</li> </ul> |
| LEARNING ASSUMED TO BE IN PLACE | Building blocks (knowledge, skill and understanding) assumed to be in place                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| LEARNING PROGRAM                | A “learning program is a set of learning activities in which the learner will be involved in working towards the achievement of one or more outcomes.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| LEVEL                           | Each unit standard is registered on the NQF at a particular level based on the complexity of the outcomes and stringency of the standards.                                                                                                                                                                                                                                                                                                                                                                                                                            |
| NQF                             | National Qualifications Framework                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| RANGE STATEMENTS                | The range statements indicate detailed requirements of the assessment criteria.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RE-ASSESSMENT        | <ul style="list-style-type: none"> <li>• The same method and instrument may be used, but, the task and materials should be different</li> <li>• The task and materials should, however, be of the same complexity and level as the previous ones.</li> <li>• In case the methods and instruments are changed it must be ensured that they are appropriate for the outcomes specified.</li> </ul> |
| RPL ASSESSMENT       | RPL is a way of recognizing what individuals already know and can do. The basic premise underlying it is that people, especially mature adults, learn many things inside and outside the formal structures of education and training and that irrespective of where, how and when learning was acquired, subject to assessment, it can be worthy of recognition and credit.                      |
| RPL                  | Recognition of prior learning                                                                                                                                                                                                                                                                                                                                                                    |
| SASSETA              | Safety & security sector education and training authority.                                                                                                                                                                                                                                                                                                                                       |
| SAQA                 | SOUTH AFRICAN QUALIFICATION AUTHORITY                                                                                                                                                                                                                                                                                                                                                            |
| SPECIFIC OUTCOMES    | <p>An outcome is a statement that describes the required competency that must be demonstrated by the learner on successful completion of a training intervention.</p> <p>Key performance areas in the standard are given, the focus is <i>not</i> only on current competence, but also on future competence in this standard.</p>                                                                |
| SUMMATIVE ASSESSMENT | Assessment used to record the level of competence achieved by a learner over a period of time, e.g. at the end of a program, term or year.                                                                                                                                                                                                                                                       |
| QUALIFICATIONS       | Represent a planned combination of learning outcomes which has a defined purpose and intended to provide learners with applied competence and a basis for further learning                                                                                                                                                                                                                       |
| UNIT STANDARDS       | the “building blocks” of qualifications. All qualifications are plotted on the National Qualifications Framework (NQF). “A statement of the outcomes that the learner must achieve to earn credit. Standards are nationally agreed and internationally comparable.”                                                                                                                              |





**Describe, apply, analyse and calculate shape and motion in 2-and 3-dimensional space in different contexts**

| SAQA US ID                                                    | UNIT STANDARD TITLE                                                                                        |                           |                      |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|---------------------------|----------------------|
| 9013                                                          | Describe, apply, analyse and calculate shape and motion in 2-and 3-dimensional space in different contexts |                           |                      |
| ORIGINATOR                                                    |                                                                                                            | REGISTERING PROVIDER      |                      |
| SGB Math Literacy, Math, Math Sciences L 2 -4                 |                                                                                                            |                           |                      |
| FIELD                                                         |                                                                                                            | SUBFIELD                  |                      |
| Field 10 - Physical, Mathematical, Computer and Life Sciences |                                                                                                            | Mathematical Sciences     |                      |
| ABET BAND                                                     | UNIT STANDARD TYPE                                                                                         | NQF LEVEL                 | CREDITS              |
| Undefined                                                     | Regular-Fundamental                                                                                        | Level 3                   | 4                    |
| REGISTRATION STATUS                                           | REGISTRATION START DATE                                                                                    | REGISTRATION END DATE     | SAQA DECISION NUMBER |
| Reregistered                                                  | 2008-01-22                                                                                                 | 2011-01-22                | SAQA 0160/05         |
| LAST DATE FOR ENROLMENT                                       |                                                                                                            | LAST DATE FOR ACHIEVEMENT |                      |
| 2012-01-22                                                    |                                                                                                            | 2015-01-22                |                      |

This unit standard does not replace any other unit standard and is not replaced by any other unit standard.

#### PURPOSE OF THE UNIT STANDARD

This unit standard is designed to provide credits towards the mathematical literacy requirements of the NQF at level 3. The essential purposes of the mathematical literacy requirements are that, as the learner progresses with confidence through the levels, the learner will grow in:

- An insightful use of mathematics in the management of the needs of everyday living to become a self-managing person
- An understanding of mathematical applications that provides insight into the learner's present and future occupational experiences and so develop into a contributing worker
- The ability to voice a critical sensitivity to the role of mathematics in a democratic society and so become a participating citizen.

People credited with this unit standard are able to:

- Measure, estimate, and calculate physical quantities in practical situations relevant to the adult in life or the workplace
- Explore describe and represent, interpret and justify geometrical relationships and conjectures to solve problems in two and three dimensional geometrical situations

#### LEARNING ASSUMED TO BE IN PLACE AND RECOGNITION OF PRIOR LEARNING

The credit value is based on the assumption that people starting to learn towards this unit standard are competent in Mathematical Literacy and Communications at NQF level 2.

#### UNIT STANDARD RANGE

**UNIT STANDARD 9013: Describe, apply, analyse and calculate shape and motion in 2-and 3-dimensional space in different contexts.**



The scope of this unit standard includes length, surface area, volume, mass, speed ; ratio and proportion; making and justifying conjectures.

Contexts relevant to the adult, the workplace and the local community.

More detailed range statements are provided for specific outcomes and assessment criteria as needed.

## **Specific Outcomes and Assessment Criteria:**

### **SPECIFIC OUTCOME 1**

Measure, estimate, and calculate physical quantities in practical situations.

#### **OUTCOME NOTES**

Measure, estimate, and calculate physical quantities in practical situations relevant to the adult in life or the workplace

#### **OUTCOME RANGE**

- Basic instruments to include those readily available such as rulers, measuring tapes, measuring cylinders or jugs, thermometers, spring or kitchen balances, watches and clocks.
- In situations which necessitate it such as in the workplace, the use of more accurate instruments such as vernier callipers, micrometer screws, stop watches and chemical balances.
- Quantities to estimate or measure to include length/distance, area, mass, time, speed and temperature.
- Estimate the area and volume of simple irregular shapes and objects.
- The quantities should range from the low or small to the high or large.
- Mass, volume temperature, distance, and speed values are used in practical situations relevant to the learner or the workplace.
- Calculations involving the effects on area and volume when altering linear dimensions.
- Calculate heights and distances using Pythagoras' theorem.
- Calculate surface areas and volumes of right prisms (i.e., end faces are polygons and the remaining faces are rectangles) and cylinders from measurements in practical situations relevant to the life of the learner or in the workplace.

### **ASSESSMENT CRITERIA**

#### **ASSESSMENT CRITERION 1**

1. Scales on the measuring instruments are read correctly.

#### **ASSESSMENT CRITERION 2**

2. Quantities are estimated to a tolerance justified in the context of the need.

#### **ASSESSMENT CRITERION 3**

3. The appropriate instrument is chosen to measure a particular quantity.

#### **ASSESSMENT CRITERION 4**

4. Quantities are measured correctly to within the least step of the instrument.

#### **ASSESSMENT CRITERION 5**

5. Calculations are carried out correctly.

#### **ASSESSMENT CRITERION 6**

6. Symbols and units are used in accordance with SI conventions and as appropriate to the situation.

### **SPECIFIC OUTCOME 2**

Explore, describe and represent, interpret and justify geometrical relationships and conjectures.

#### **OUTCOME NOTES**

Explore, describe and represent, interpret and justify geometrical relationships and conjectures to solve problems in two and three dimensional geometrical situations.

#### **OUTCOME RANGE**

- Applications taken from different contexts such as packaging, arts, building construction, dressmaking.

- The use of tessellations and symmetry in artifacts and in architecture.
- Use rough sketches to interpret, represent and describe situations.
- Use and interpret scale drawings of plans (e.g., plans of houses or factories; technical diagrams of simple mechanical household or work related devices such as jacks,
- Nets of prisms and cylinders.
- Road maps relevant to the local community.
- The use of the Cartesian co-ordinate system in determining location and describing relationships in at least two dimensions.

**ASSESSMENT CRITERIA****ASSESSMENT CRITERION 1**

1. Descriptions are based on a systematic analysis of the shapes and reflect the properties of the shapes accurately, clearly and completely.

**ASSESSMENT CRITERION 2**

2. Descriptions include quantitative information appropriate to the situation and need.

**ASSESSMENT CRITERION 3**

3. Conjectures as appropriate to the situation, are based on well-planned investigations of geometrical properties.

**ASSESSMENT CRITERION 4**

4. Representations of the problems are consistent with and appropriate to the problem context. The problems are represented comprehensively and in mathematical terms.

**ASSESSMENT CRITERION 5**

5. Results are achieved through efficient and correct analysis and manipulation of representations.

**ASSESSMENT CRITERION 6**

6. Problem-solving methods are presented clearly, logically and in mathematical terms.

**ASSESSMENT CRITERION 7**

7. Solutions are correct and are interpreted and validated in terms of the context of the problem.

**UNIT STANDARD ACCREDITATION AND MODERATION OPTIONS**

Accreditation Option: Providers of learning towards this unit standard will need to meet the accreditation requirements of the GENFETQA.

Moderation Option: The moderation requirements of the GENFETQA must be met in order to award credit to learners for this unit standard.

**UNIT STANDARD ESSENTIAL EMBEDDED KNOWLEDGE**

The following essential embedded knowledge will be assessed through assessment of the specific outcomes in terms of the stipulated assessment criteria. Candidates are unlikely to achieve all the specific outcomes, to the standards described in the assessment criteria, without knowledge of the listed embedded knowledge. This means that the possession or lack of the knowledge can be inferred directly from the quality of the candidate's performance against the standards.

- Properties of geometric shapes
- Length, area, volume, mass, time, temperature, speed
- The Cartesian system
- Scale drawing

**Critical Cross-field Outcomes (CCFO):**

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**UNIT STANDARD 9013: Describe, apply, analyse and calculate shape and motion in 2-and 3-dimensional space in different contexts.**



**UNIT STANDARD CCFO IDENTIFYING**

- Identify and solve problems using critical and creative thinking:  
Solve a variety of problems involving space, shape and time using geometrical techniques related to the life or workplace of the learner

**UNIT STANDARD CCFO ORGANISING**

- Collect, analyse, organise and critically evaluate information:  
Gather, organise, and interpret information about objects and processes.

**UNIT STANDARD CCFO COMMUNICATING**

- Communicate effectively:  
Use everyday language and mathematical language to describe properties, processes and problem solving methods.

**UNIT STANDARD CCFO SCIENCE**

- Use mathematics:  
Use mathematics to analyse, describe and represent realistic and abstract situations and to solve problems relevant to the adult, the workplace and the local community.

**QUALIFICATIONS UTILISING THIS UNIT STANDARD:**

|             | ID                    | QUALIFICATION TITLE                                                   | LEVEL   | STATUS                                        | END DATE   |
|-------------|-----------------------|-----------------------------------------------------------------------|---------|-----------------------------------------------|------------|
| Fundamental | <a href="#">48794</a> | National Certificate in Quality Checking of Tyres and Tyre Components | Level 3 | Reregistered                                  | 2010-07-17 |
| Fundamental | <a href="#">48902</a> | National Certificate: Abattoir Supervision                            | Level 3 | Reregistered                                  | 2010-05-15 |
| Fundamental | <a href="#">58011</a> | National Certificate: Acoustic Piano Back-framing                     | Level 3 | Registered                                    | 2010-03-12 |
| Fundamental | <a href="#">14892</a> | National Certificate: Agricultural Machinery Technician               | Level 3 | Reregistered                                  | 2008-09-13 |
| Fundamental | <a href="#">20720</a> | National Certificate: Air-Conditioning, Refrigeration and Ventilation | Level 3 | Passed the End Date - Status was "Registered" | 2004-06-13 |
| Fundamental | <a href="#">48963</a> | National Certificate: Air-Conditioning, Refrigeration and Ventilation | Level 3 | Reregistered                                  | 2010-09-18 |
| Fundamental | <a href="#">49048</a> | National Certificate: Animal Production                               | Level 3 | Reregistered                                  | 2010-08-07 |
| Fundamental | <a href="#">49772</a> | National Certificate: Archaeology                                     | Level 3 | Registered                                    | 2008-11-30 |
| Fundamental | <a href="#">48864</a> | National Certificate: Auctioneering Support                           | Level 3 | Reregistered                                  | 2010-10-30 |
| Fundamental | <a href="#">21005</a> | National Certificate: Automotive Component Manufacturing and Assembly | Level 3 | Reregistered                                  | 2008-11-01 |
| Fundamental | <a href="#">58497</a> | National Certificate: Automotive Repair and Maintenance               | Level 3 | Registered                                    | 2011-02-06 |
| Fundamental | <a href="#">22859</a> | National Certificate: Autotronics                                     | Level 3 | Reregistered                                  | 2009-11-07 |
| Fundamental | <a href="#">49993</a> | National Certificate: Auxiliary Nursing                               | Level 3 | Registered                                    | 2008-11-30 |
| Fundamental | <a href="#">20184</a> | National Certificate: Banking                                         | Level 3 | Reregistered                                  | 2009-01-10 |
| Fundamental | <a href="#">48745</a> | National Certificate: Beauty Technology                               | Level 3 | Reregistered                                  | 2010-10-30 |
| Fundamental | <a href="#">58375</a> | National Certificate: Bookkeeping                                     | Level 3 | Registered                                    | 2010-09-12 |
|             | <a href="#">50308</a> | National Certificate: Bread and Flour                                 |         |                                               | 2009-      |